

LESSON 6-3 • SESSION 1 • PREREQUISITE PRACTICE A • SKILL 1 OF 2

Repeated Addition vs. Multiplication

Builds on Grade 3 • 3.OA.A.1

TODAY'S LESSONYou will write $5 \times 5 \times 5 \times 5$ as a power. It is not the same as $5 + 5 + 5 + 5$.**YOU NEED TO**Find the value of a repeated sum and a repeated product, like $3 + 3 + 3$ and $3 \times 3 \times 3$. **You will use it in Practice Form A, problems 3 and 5.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Count** how many times the number repeats.
- 2** For repeated **+**, multiply the number by the count.
- 3** For repeated **$\times$** , multiply the number by itself that many times.

WATCH ME*the steps in action***Find $3 + 3 + 3 + 3$ and $3 \times 3 \times 3 \times 3$.****Step 1** 3 repeats 4 times in each.**Step 2** $3 + 3 + 3 + 3 = 3 \times 4 = 12$.**Step 3** $3 \times 3 \times 3 \times 3 = 9 \times 3 \times 3 = 81$.**Answer 12 and 81. They are not the same.****TRY IT WITH HELP***fill in each step*

- 1.**
- Find
- $2 + 2 + 2$
- and
- $2 \times 2 \times 2$
- .

Step 1 2 repeats _____ times.**Step 2** $2 + 2 + 2 = 2 \times 3 =$ _____**Step 3** $2 \times 2 \times 2 = 4 \times 2 =$ _____

- 2.**
- Find
- $5 + 5 + 5$
- and
- $5 \times 5 \times 5$
- .

Step 1 5 repeats _____ times.**Step 2** $5 + 5 + 5 = 5 \times 3 =$ _____**Step 3** $5 \times 5 \times 5 = 25 \times 5 =$ _____**YOUR TURN***use the steps on your own*

- 3.**
- Find each value.

a. $4 + 4 + 4 =$ _____

b. $4 \times 4 \times 4 =$ _____

c. $10 + 10 =$ _____

d. $10 \times 10 =$ _____

e. $3 + 3 + 3 =$ _____

f. $3 \times 3 \times 3 =$ _____

Name Factors and Products

Builds on Grade 4 • 4.OA.B.4

TODAY'S LESSONYou will write $1/2 \times 1/2 \times 1/2$ as a power. It has 3 factors of $1/2$.**YOU NEED TO**Name and count the factors in a product like $7 \times 7 \times 7$. **You will use it in Practice Form A, problems 6 and 8.****STEPS TO FOLLOW***use these same steps on every problem*

- 1** The numbers being multiplied are the **factors**.
- 2** **Count** the factors.
- 3** **Multiply** to find the product.

WATCH ME*the steps in action***Name the factors in $6 \times 6 \times 6$. Find the product.****Step 1** The factors are 6, 6 and 6.**Step 2** There are 3 factors.**Step 3** $6 \times 6 \times 6 = 36 \times 6 = 216$.**Answer** **3 factors of 6. The product is 216.****TRY IT WITH HELP***fill in each step***4.** Name the factors in 4×4 . Find the product.**Step 1** The factors are _____ and _____.**Step 2** There are _____ factors.**Step 3** $4 \times 4 =$ _____**5.** Name the factors in $1/2 \times 1/2 \times 1/2$. Find the product.**Step 1** The factors are _____, _____ and _____.**Step 2** There are _____ factors.**Step 3** $1/2 \times 1/2 \times 1/2 =$ _____**YOUR TURN***use the steps on your own***6.** How many factors? Find the product.**a.** $2 \times 2 \times 2 \times 2$ _____**b.** $10 \times 10 \times 10$ _____**c.** 7×7 _____**d.** $3 \times 3 \times 3$ _____**READY CHECK***a problem like the one in today's lesson***7.** Rafi wrote $8 \times 8 \times 8$ as 8^3 . How many factors of 8 are in $8 \times 8 \times 8$?**A** 8**B** 3**C** 24**D** 512**CHECK YOURSELF**
☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #____